ASSIGNMENT 3

ARYAN

September 28, 2026

Que 1

Write LaTeX code to produce the following expression:

$$\overbrace{a+b+c}^{\text{First Group}}+\underbrace{x+y+z}_{\text{second Group}}=\overbrace{\underbrace{a+b}_{\text{Sum 1}}+c}^{\text{Total}}$$

Que 2

Create the following array using the array environment:

$$A=\left[\begin{array}{c|ccc} & x_1 & x_2 & x_3 \\ \hline y_1 & a_{11} & a_{12} & a_{13} \\ y_2 & a_{21} & a_{22} & a_{23} \\ y_3 & a_{31} & a_{32} & a_{33} \end{array}\right]$$

Que 3

Write the following piecewise function using the cases environment:

$$f(x)=\begin{cases} x^2+2x+1, & x<0, \\ \sqrt{x}+1, & 0\le x<4, \\ \dfrac{x^2-1}{x-1}, & x\ge 4. \end{cases}$$

Que 4

Typeset the following matrix:

$$\left[\begin{array}{ccccc} \overbrace{\phantom{a_{11}\ a_{12}\ a_{13}}}^{\text{Column Group 1}} & & & \overbrace{\phantom{a_{14}\ a_{15}}}^{\text{Column Group 2}} & \\ a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \end{array}\right]$$

Que 5

Create the following expression

$$F(x) = \overbrace{\sum_{i=1}^{n}(a_i x_i^2 + b_i x_i + c_i)}^{\text{Quadratic Contribution}} + \overbrace{\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} A \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}^{\text{Matrix Contribution}}$$

$$\text{Total Energy}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}.$$

Que 6

Create the following table with proper alignment, borders, and headings.

Roll No.	Name	Department	Semester	CGPA
101	Amit	CSE	2	8.45
102	Neha	CSE	2	9.12
103	Rahul	ECE	4	8.76
104	Priya	ECE	4	9.01

Que 7

Create the following table.

Academic Performance		
Student	Midterm	Final
A	78	89
B	85	92
C	91	94

Que 8

Create the following table.

Department	Course	Credits	Grades
CSE	Data Structures	4	A
	Machine Learning	4	A+
	Operating Systems	3	B+
ECE	Digital Electronics	4	A
	Signals	3	B+

Que 9

Create the following scientific table.

Model	Parameters	FLOPs	Accuracy	Energy (kWh)
ResNet18	11.7M	1.8G	94.2%	0.00093
MobileNetV2	3.5M	0.3G	92.8%	0.00061
EfficientNet-B0	5.3M	0.39G	93.5%	0.00072

Que 10

Create a student-performance table.

Student	Mathematics	Physics	Computer Science
Amit	92	81	95
Neha	72	88	91
Rahul	68	74	78
Priya	96	94	98

Que 11

Create a table containing paragraphs inside cells.

Technique	Description	Advantages	Limitations
Pruning	Removes unnecessary parameters from a neural network.	Reduces model size and computation.	May reduce accuracy.
Quantization	Represents weights using lower precision.	Reduces memory and energy consumption.	Can introduce numerical error.
Knowledge Distillation	Transfers knowledge from a large teacher model to a smaller student model.	Produces compact models.	Requires training a teacher model.

Que 12

Create a professional table using the booktabs package.

Model	Parameters	FLOPs	Accuracy	Energy (kWh)
ResNet18	11.7M	1.8G	94.2%	0.00093
MobileNetV2	3.5M	0.3G	92.8%	0.00061
EfficientNet-B0	5.3M	0.39G	93.5%	0.00072

Table 1: Caption

Que 13

Create a table where one cell contains another table.

<table>
<tr><th>Model</th><th>Architecture</th><th colspan="2">Performance</th></tr>
<tr><td rowspan="3">CNN</td><td rowspan="3">Conv → ReLU → Pool</td><td>Metric</td><td>Value</td></tr>
<tr><td>Accuracy</td><td>94 %</td></tr>
<tr><td>F1 Score</td><td>94 %</td></tr>
<tr><td rowspan="3">ResNet</td><td rowspan="3">Conv → BN → ReLU</td><td>Metric</td><td>Value</td></tr>
<tr><td>Accuracy</td><td>96 %</td></tr>
<tr><td>F1 Score</td><td>95 %</td></tr>
<tr><td rowspan="3">ViT</td><td rowspan="3">Attention → MLP</td><td>Metric</td><td>Value</td></tr>
<tr><td>Accuracy</td><td>97 %</td></tr>
<tr><td>F1 Score</td><td>96 %</td></tr>
</table>

Que 14

Create the following table containing mathematical expressions.

Model	Loss Function	Parameters	Complexity
Linear	$L = \frac{1}{n}\sum_{i=1}^{n}(y - \hat{y}_i)^2$	10K	$O(n)$
CNN	$L = -\sum y \log \hat{y}_i$	1.2M	$O(n^2)$
Transformer	$L = -\sum_i y_i \log p_i$	110M	$O(n^2 d)$